# PE&RC PhD PROJECT PROPOSAL

December 27, 2025

## 1 General Information

### 1.1 ABOUT THE PROJECT PROPOSAL AND EVALUATION CRITERIA

The PE&RC PhD proposal is written by the PhD candidate in consultation with and feedback from the supervisory team. After submission, the proposal is sent out for review. The candidate and the supervisory team are encouraged to suggest possible independent reviewers that can provide constructive feedback to the candidate.
The reviewers are asked to evaluate the proposal on:

- *Scientific quality*

  Clarity of writing, depth of literature review, definition of research question(s) and objective(s), and originality of subject and/or methodology.

- *Feasibility*

  Feasibility given the 4-year time frame, (financial) resources, effectiveness of the proposed approach/strategy, logistical/practical aspects and evaluation of the expertise within the supervising team and/or collaborations for adequate support in the proposed project.

- *Scientific and/or societal impact*

  The importance of the potential research results in the short and long term in the field of research, potential use and relevance in other scientific disciplines, and potential use and relevance for society (economic, technical, social, cultural), including technological application.

### 1.2 GENERAL PROJECT INFORMATION:

| | |
|---|---|
| Main PE&RC affiliated Institute/University | Wageningen University |
| Main PE&RC research group | |
| Other PE&RC research groups involved | |
| Where will the research be conducted (country) | Netherlands |
| Full name of the PhD candidate | Lisa Boterman |
| Project duration | FROM 01/02/2025 TO 31/01/2029 |
| Time spent on PhD project | Fulltime |
| Principal supervisor/Promotor | Veerle Vanacker |
| Additional supervisor(s) | Jasper Candel & Annegret Larsen |

### 1.3 ETHICS

*Ethical issues such as experiments with animals (vertebrates), work with genetically modified organisms, work with material under the Nagoya protocol, or personal data privacy considerations (AVG) may have legal implications. Also, ethical guidelines of the respective research institution may apply.*

| | |
|---|---|
| Do any of the above mentioned, or other ethical issues apply to this project? | NO |
| If YES, please elaborate: | |

## 1.4 PE&RC RESEARCH THEME(S)

*In which of the PE&RC research theme(s) does the project fit (multiple options possible). Please elaborate.*

| **Themes** | **Elaboration** |
|---|---|
| ✓ Biodiversity and Systems Dynamics<br>❏ One Health<br>❏ Redesign of (agro-) ecological systems<br>❏ Data and Engineering Science for Life and Environment | This project will contribute to the PE&RC mission by improving our understanding of *the functioning of natural ecosystems* (rivers) towards the *provision of ecosystem services* and *protection, restoration and sustainable use of these ecosystems*. The proposed research will encompass *various spatial and temporal scales*, and will endeavour to *develop quantitative approaches* in assessing carbon sequestration in these systems. |

# 2 Proposal Structure

## 2.1 TITLE:

**Dynamics of carbon storage: the case for river floodplains**

## 2.2 ABSTRACT (max. 250 words):

Rivers profoundly impact the partitioning of the terrestrial carbon pool by transporting and processing quantities of carbon disproportionate to their surface area. Depositional environments such as floodplains therefore potentially store significant quantities of carbon. Controls and processes facilitating carbon accumulation and storage, however, are poorly constrained, and constitute a significant knowledge gap in the context of carbon budgeting of river networks. This research aims to address this gap by assessing the spatial and temporal variability in carbon retention and preservation in floodplain sediments. This will be achieved by investigating several controls on potential carbon accumulation and storage: (1) variation in regional hydrology; (2) beaver presence; and (3) landslide sediment inputs; representing process domains in lower, middle and upper river reaches respectively. Carbon storage will be quantified relative to the temporal and spatial scales of the processes and feedbacks that drive accumulation of sediments and biomass. Subsequently, the stability of carbon stored in these depositional systems will be assessed to further constrain variability in floodplain carbon stocks through time. A fundamental understanding of the controls and processes enhancing and maintaining carbon storage, and their associated time scales, will enable better constraint on carbon budgets in inherently dynamic river systems, and better inform climate change mitigation and restoration measures.

## 2.3 RESEARCH TOPIC AND BACKGROUND

Rivers are increasingly being recognised as contributing substantially to the global carbon (C) cycle [3, 16, 5]. The terrestrial carbon pool is profoundly impacted by rivers, which facilitate stream metabolism, atmospheric gas exchange and storage of organic matter in riparian biomass and in floodplain sediments [3, 5, 94]. Compared to terrestrial ecosystems where respiration, storage and export of organic carbon is limited by gross primary production [16], the amount of carbon transported by rivers can be of the same magnitude as the carbon balance of their entire catchment [3]. Estimates suggest that 0.6 Pg C $yr^1$ of the 5.1 Pg C $yr^1$ [21] entering inland waters is stored within these systems [70]. However, this has been recognised to be a conservative estimate with a high uncertainty [21], with some studies suggesting potential C burial in inland waters to be as high as 1.5 Pg C $yr^1$ [3] to 4.2 Pg C $yr^1$ [49]. This implies significant re-deposition and reburial of sediments en route to the oceans. Floodplains, as depositional environments, therefore can potentially store significant amounts of carbon relative to their adjacent uplands [93, 71, 80, 22].

However, carbon 'storage' in the river or floodplain context, is relatively transient when compared to, for example, the geologic record. Ultimately the functioning of floodplains in attenuating fluxes of water and sediment only delays downstream transport [92]. Changes in the characteristics of stored carbon through physical, chemical or biological processes can result in C stocks being highly vulnerable to transport and mineralisation, and thus variable in time and space [85, 73, 72, 76, 92]. The quantification of carbon storage is further complicated by the potential for mobilisation, or reintroduction of 'aged' carbon into the system (e.g. [17, 1, 30]). Storage should then be considered relative to the timescales of the controls, processes and feedbacks driving accumulation, preservation, and potential remobilisation. Limited quantification of these factors, and their respective spatial and temporal scales, impedes our understanding of carbon cycling in river networks [5, 37, 94]. In particular, small watersheds and mountainous rivers (which can contribute disproportionately to carbon fluxes [34]) are associated with higher uncertainties in carbon storage and export, and warrant further research [21].

In its unaltered state, a river consists of its water-bearing channel(s), and its riparian zone (Fig. 1), defined here as encompassing floodplains, the hyporheic zone, and depositional landforms such as

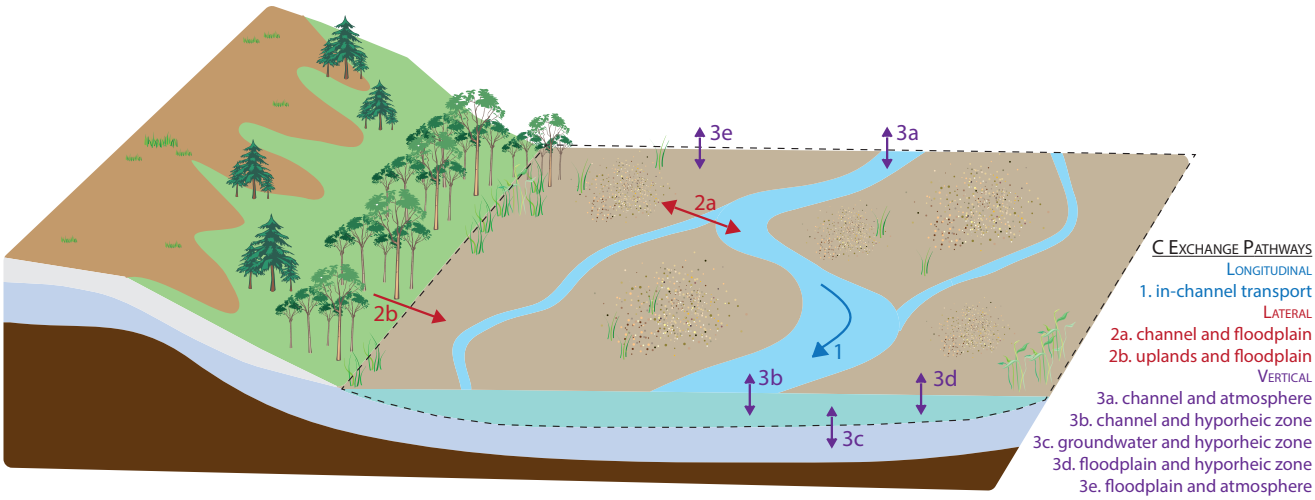


**Figure 1:** Conceptual illustration of a river and C exchange pathways. Black dashed line delineates the 'river' as defined for the purpose of this study, comprised of the channel, floodplain and (potential) hyporheic zone. Carbon (C) exchange pathways are shown by arrows and listed in colours corresponding to longitudinal (blue), lateral (red) and vertical (purple) fluxes.

alluvial fans after Wohl et al. [94]. Carbon is cycled through this system through various pathways as organic and inorganic carbon, both in particulate and dissolved form [70]. Organic carbon (OC) is biomass-derived and can be autochthonous (associated with primary production) or allochthonous (sourced from adjacent uplands or transported from upstream; [81]). Inorganic carbon (IC) in particulate form is derived from sediments, mineral weathering processes and carbonate precipitates, while dissolved inorganic carbon (DIC) originates from mineral dissolution into various carbon species ($CO_2, H_2CO_3, HCO_3^-, CO_3^{2-}$), surface water $CO_2$ exchange, or as the product of biological respiration and mineralisation of organic matter [3]. The pathways in which these various forms of carbon is cycled through and within rivers are illustrated in Fig. 1. Carbon storage in floodplains can be facilitated through various processes. These include overbank deposition and resulting floodplain aggradation [63, 81, 92] or burial of organic matter (OM) or organic rich soil horizons [27, 30, 33], sequestration of OC produced in-situ through pedogenesis [87, 97] or peat formation [58, 8], or through damming of the channel and lake formation resulting in sedimentation [62, 93, 90] or accumulation of biogenic carbon [79]. These processes often take place in association with each other.

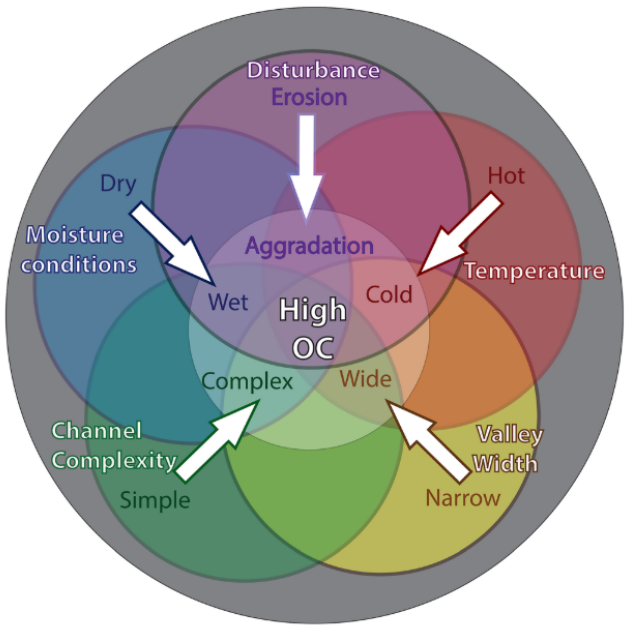


**Figure 2:** Conceptual diagram for optimal conditions for organic carbon retention (towards centre of diagram). From [81].

The spatial distribution of carbon within a river system at the landscape scale is largely determined by geology - with uplift and lithology affecting weathering rates, sediment production and nutrient availability - and climate - regulating disturbance and runoff, primary productivity, and decomposition [81, 60]. These governing factors and their relationships at various scales determine the conditions that enable retention of carbon in river systems (Fig. 2). Locally, carbon

retention and preservation are controlled by geomorphic and biotic factors, occuring primarily in relatively wide, low gradient river segments (floodplains) [91, 81, 82, 95]). Segments that are characterised by retention are often present throughout the river network where a local base level control forces a reduction in channel gradient [54] and re-adjustment of the longitudinal profile (Fig. 3) towards equilibrium state [57]. This readjustment can result in a variety of depositional processes (as mentioned above) facilitating carbon storage both upstream and downstream of the control on local base level.

Controls on local base levels are often linked to distinct process domains within the river system [60]. For example, upstream reaches are more likely to be affected by landsliding, debris flow or logjams (e.g.[93, 33]), while mid to downstream reaches are more susceptible to biotic influences (e.g. [51, 29], flooding-induced channel migration and backswamp formation [60], or in low-energy settings, the formation of organic floodplains [9], or self-constrained channels with heterogeneous floodplains [10]. Although it has been recognised that these various controls contribute to potential sites for carbon storage, this storage is poorly constrained [21], both spatially between river segments and networks [94], and temporally in terms of the persistence (physical preservation and chemical degradation) of carbon [76, 72] once retained in floodplain sediments [92].

In the context of carbon budgeting or nature restoration, growing recognition of ecosystem services such as carbon storage has seen river management practices increasingly reverting toward restoration of the inherent spatial heterogeneity of river systems [7, 95, 14]. However, the dynamic nature of such systems do not lend themselves to our short-term, and often static, politically-driven perspectives on river management. A fundamental understanding of the controls and processes enhancing and maintaining carbon storage, and their associated time scales, is therefore required in order to better constrain carbon budgets in river systems [3, 5, 37] and inform sustainable restoration measures.

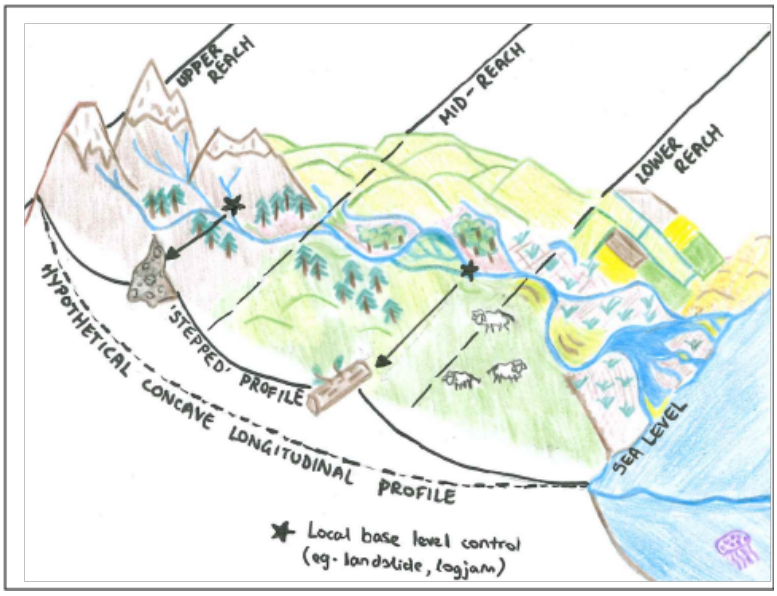


**Figure 3:** Schematic illustration of a river network, showing hypothetical concave profile [57] from upper reach to sea level, and 'stepped' profile formed by the presence of controls creating local base levels (e.g. landslide, logjams).

## AIM AND OBJECTIVE

Various depositional controls can influence channel gradients and sedimentation regimes, thus facilitating carbon transfer, accumulation and storage in river floodplains. We hypothesise that local depositional controls are important drivers in floodplain carbon accumulation and storage, and in the creation of potential carbon storage. However, their magnitude is presently poorly constrained, both spatially within different floodplain settings, and temporally in terms of the persistence of accumulated carbon. This constitutes a significant knowledge gap in the context of carbon budgeting within river networks. This research aims **to assess the spatial and temporal variability in carbon accumulation and storage in floodplain sediments under local and regional scale controls on deposition**. This project will address this knowledge gap in four chapters. Chapters 1-3 will investigate the controls of regional hydrology (sea level rise and precipitation regime), beaver presence, and landsliding - representing process domains in lower, middle and upper river reaches respectively - on inputs and retention of sediment and/or carbon in these reaches. Chapter 4 will investigate the chemical stability of floodplain carbon across these different depositional

environments. An understanding of the fundamental processes creating and sustaining potential carbon storage throughout the river network will enable better constraint on local and regional carbon budgets, and provide a foundation for upscaling these to similar fluvial environments elsewhere.

## 2.4 SCIENTIFIC AND SOCIETAL IMPACT

Global initiatives such as the Paris Agreement [86] and the IPCC report [39] stress the importance of mitigation measures to reduce climate change impacts such as floods, droughts, wildfires and biodiversity loss. Rivers have been recognised to provide essential ecosystem services such as water retention, habitat provision and connectivity, and carbon sequestration, with nature restoration goals being cemented into policy frameworks such as the UN Sustainable Development Goals (Goal 15.1, 15.3; [64] and the EU nature restoration law [68]. This research will contribute to our fundamental understanding of river systems by providing valuable insights into hydrological, biotic, and geomorphic controls on floodplain carbon dynamics, that can be used to inform successful nature-based approaches to river restoration and climate change mitigation. Assessment of both the physical preservation and chemical stability of floodplain C will provide perspective on how floodplain carbon stocks evolve over time; this perspective is essential to consider for the sustainable management of systems as inherently dynamic systems such as rivers.

This research will also produce insights towards better constraint on terrestrial carbon budgets by assessing floodplain environments (and their C contents) as distinct landforms. Present estimates of soil carbon stocks do not distinguish floodplains as separate units to their adjacent uplands with different soil carbon contents[81], despite increasing evidence for their disproportionate contribution to terrestrial C cycling [93, 71, 80, 22]. This contributes to the present high uncertainties associated with riverine C fluxes [5, 16, 21]. On more regional and local scales, the proposed chapters will provide a more comprehensive assessment of benefits and impacts associated with phenomena that potentially conflict with anthropogenic land use, such a beaver dam-induced flooding [18], or landslide-induced avulsion [43]. These insights will improve our understanding of river system dynamics to contribute towards balancing river management outcomes with nature restoration measures.

## 2.5 DESCRIPTION OF THE RESEARCH PLAN

### Chapter 1: Lower reach controls on C: Regional hydrological variation

Peatland carbon comprises 25% of terrestrial carbon stocks [96]. Regeneration efforts to restore peat often overlook floodplain settings, despite their ability to support substantial peat formation (e.g. [83, 8, 66]). In light of increasing climate change mitigation and nature restoration efforts to restore ecosystem services provided by peatlands and rivers, an understanding of peat-river interactions and their implications for carbon sequestration, is warranted. Peat formation relies on specific environmental (namely hydrological) conditions (e.g. Fig. 2) to prevent decomposition, and enable long-term preservation of biomass [12]. However, these environmental conditions exhibit considerable variability through time, and potentially more so in the context of inherently dynamic riverine settings. Water table fluctuations, vegetation type, and decomposition of OM through time all impact peat formation and its mineralisation [89], and therefore quantification of C sequestration in peat is more complex than simply assessing the depth of deposits (e.g. [67]). This chapter aims to **discern the effects of variability in hydrological controls on C content in floodplain peats**.

North-western Europe experienced large-scale peat growth throughout the Holocene [58, 89, 2]. Previous studies of rivers in the Netherlands have ascertained peat growth to be driven by both sea level rise (SLR; [15]) and/or regional precipitation and vegetation changes [58]. The Drentsche Aa valley, in the north-east of the Netherlands, experienced peat growth throughout the Holocene under various hydrological controls, with peat depths of up to 7m being preserved in the valley fill ([58]). Given the state of preservation of the Drentsche Aa (both morphologically- with the stream never having been channelised, and temporally- with deposits spanning the entire Holocene; [9]), and the extensive

information available from previous research, this site will be used to investigate the influence of hydrological controls on the C content of the peat valley-fill.

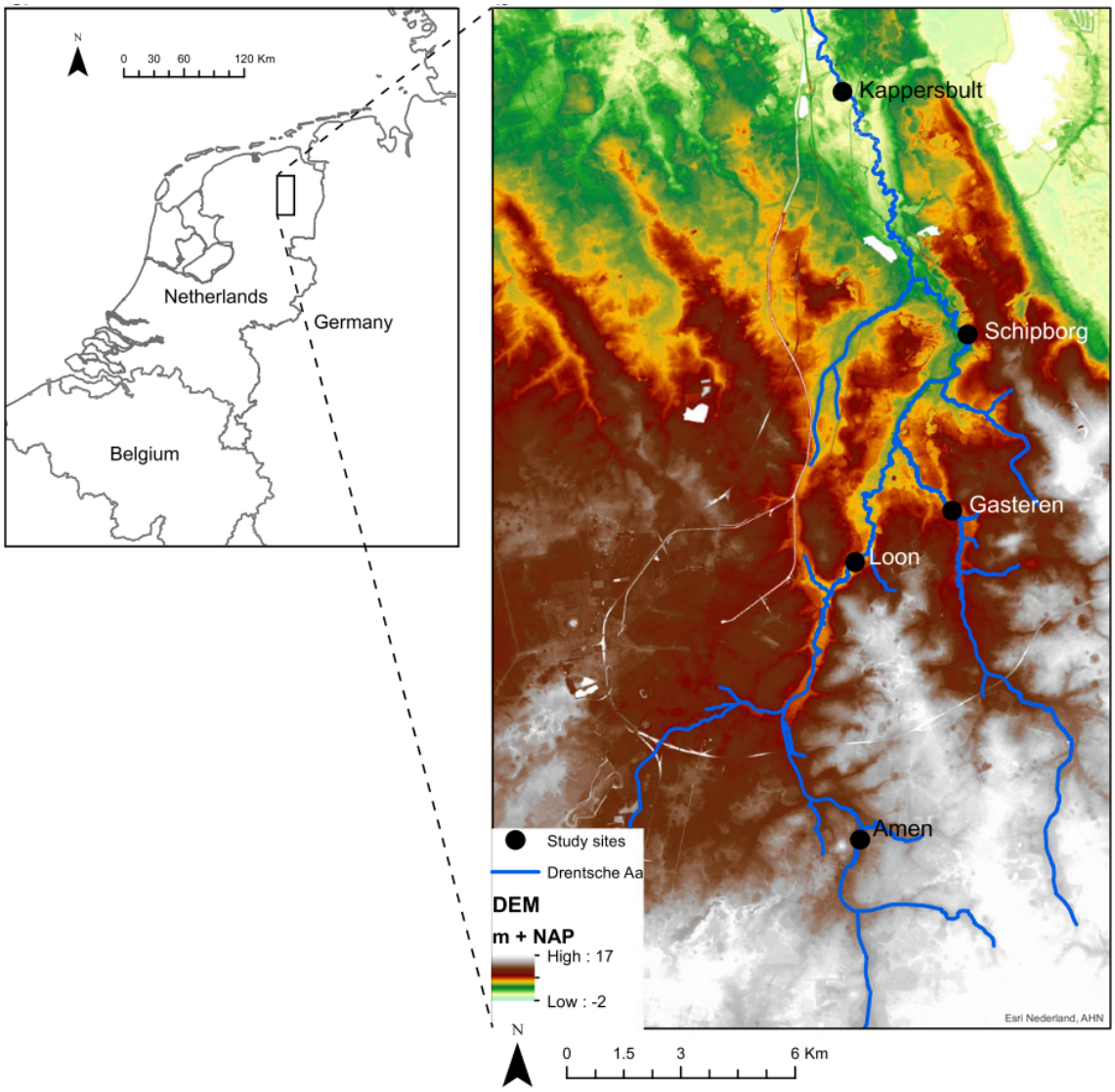


**Figure 4:** Location of Drenthsche Aa within the Netherlands and DEM of catchment with locations of study sites previously investigated byMakaske and Maas [58] and Candel et al. [9]. Further investigation at Amen, Loon and Kappersbult are proposed for this research. From Candel et al. [9].

Makaske and Maas [58] identified various hydrological controls throughout the Holocene that were reflected in different rates of peat growth in different parts of the catchment through time. In the early Holocene (11,700 - 8326 BP; [88]), peat growth rates were higher in the upper parts of the catchment, while in the late-Holocene (from 4250 BP; [88]) peat growth rates were higher in the lower catchment. This was attributed to an increase in precipitation from the onset of the Holocene, and the relative influence of continuous SLR, respectively. Mid-Holocene (8326 - 4250 BP; [88]) peat formation was attributed to a vegetation transition from coniferous to broadleaf forests (evapotranspiration 700 $mm\,yr^{-1}$ and 400 $mm\,yr^{-1}$ respectively), enabling continued rising water tables and peat growth, albeit slower than in the early Holocene. Peat growth is assumed to have stopped around the Middle Ages under valley reclamation, and the area has now been under restoration for almost 40 years [55].

Field sampling of peat valley-fill will be informed by existing data from Makaske and Maas [58] and Candel et al. [9] for sites in the upper (Amen), middle (Loon) and lower (Kappersbult) reaches (Fig. 4). C contents of peats will be measured using Rock-Eval pyrolysis (see Chapter 4). Cores of peat valley-fill will be taken in transects to enable interpretation of location relative to the valley cross-

section, and in close proximity to those obtained by Makaske and Maas [58] so that age control obtained in this study remains representative. Lithological descriptions of each core will be made, including any recognisable vegetation types and visual assessment of degree of decomposition (Makaske and Maas [58] observed strong degradation of peat in top 50 cm for Amen and Loon sites, and some decomposition of lower sections at the Amen and Kappersbult sites). Several samples will also be taken of (incipient) peat at restored sites; age constraint for these samples is provided by the start of restoration projects (within last 40 years). Subsamples will be taken to determine C content (intervals to be determined by observed lithological variation within the core) using Rock-Eval pyrolysis. Fig. 5 gives an overview of the data requirements and availability that will be used to interpret C content. Where required (and if possible), additional basal peat samples will be taken for radiocarbon dating to better constrain peat accumulation rates for the upper and middle reaches.

Existing age control and the resulting accumulation rates calculated by Makaske and Maas [58], as well as any addition radiocarbon dates obtained, will be used to classify units within peat cores by the dominant control (see above) active during their formation (early Holocene: increased precipitation; mid-Holocene: vegetation transition; late Holocene: SLR). Variation within the peat cores will also be compared to available data on paleo-climate and vegetation assemblages (e.g. regional pollen archives) to attempt to isolate the effects of regional hydrology on carbon accumulation and preservation from other, more localised controls. C contents for valley-fill peats will be analysed to determine if there are any significant differences in C accumulation per unit per individual site, as well per unit collectively across the three sites. C contents of peat under restoration will also be analysed and used to investigate how present and past C accumulation rates can be linked in order to shed light on the transformation of floodplain peat C through time.

| | | Site | | |
|---|---|---|---|---|
| **Data** | **Period** | *Lower (KB)* | *Middle (L)* | *Upper (A)* |
| Age control / growth rate | *Early HC* | | | |
| | *Mid-HC* | | | |
| | *Late HC* | | | |
| | *Present* | (site to be determined) | | |
| Lithological descriptions | | | | |
| Rock-Eval | | | | |

**Figure 5:** Overview of data to be used and availability (green= available; orange= available but sample-specific data to be acquired; red= to be acquired). KB= Kappersbult; L= Loon; A= Amen (see Fig. 4); HC= Holocene.

### Chapter 2: Mid-reach controls on C: Beaver presence

Beavers (*Castor fiber, Castor canadensis*) modify river corridors through dam construction, thereby altering the local hydrology, geomorphology and biogeochemistry of their habitat [51]. This can result in long-term carbon storage through the modification of environmental conditions enabling retention (2), such as the creation of anaerobic conditions reducing decomposition, the trapping of allochthonous organic carbon and sediments, or an increase in autochthonous carbon inputs from aquatic and riparian net ecosystem productivity [62, 28, 69, 51]. As well as providing ecosystem services, however, reintroduction and natural spread of beavers since the reduction of Eurasian populations to c. 1200 individuals a century ago [32] has also increased conflict with anthropogenic land use [18]. Therefore, river management approaches require comprehensive knowledge on beaver dam impacts. This chapter aims to **postdict the European-wide loss of floodplain carbon storage resulting from beaver eradication throughout the Holocene** to highlight the potential impact of continental-scale beaver reintroduction as a nature-based solution in the carbon sequestration and climate change mitigation context. This will be achieved using a threefold approach as detailed in Fig. 6 and described below.

1. Archaeological findings of beaver remains [50], current beaver distribution [26], and Holocene anthropogenic biomes (anthromes) and population data [41, 23], will be used to (a) estimate palaeodistribution of beavers throughout the Holocene based on findings and their respective ages for archaeological time periods ranging from the start of Mesolithic (10,800 BCE) to the year 2000 CE of the Modern period; and (b) identify potential human related patterns in eradication by analysing the

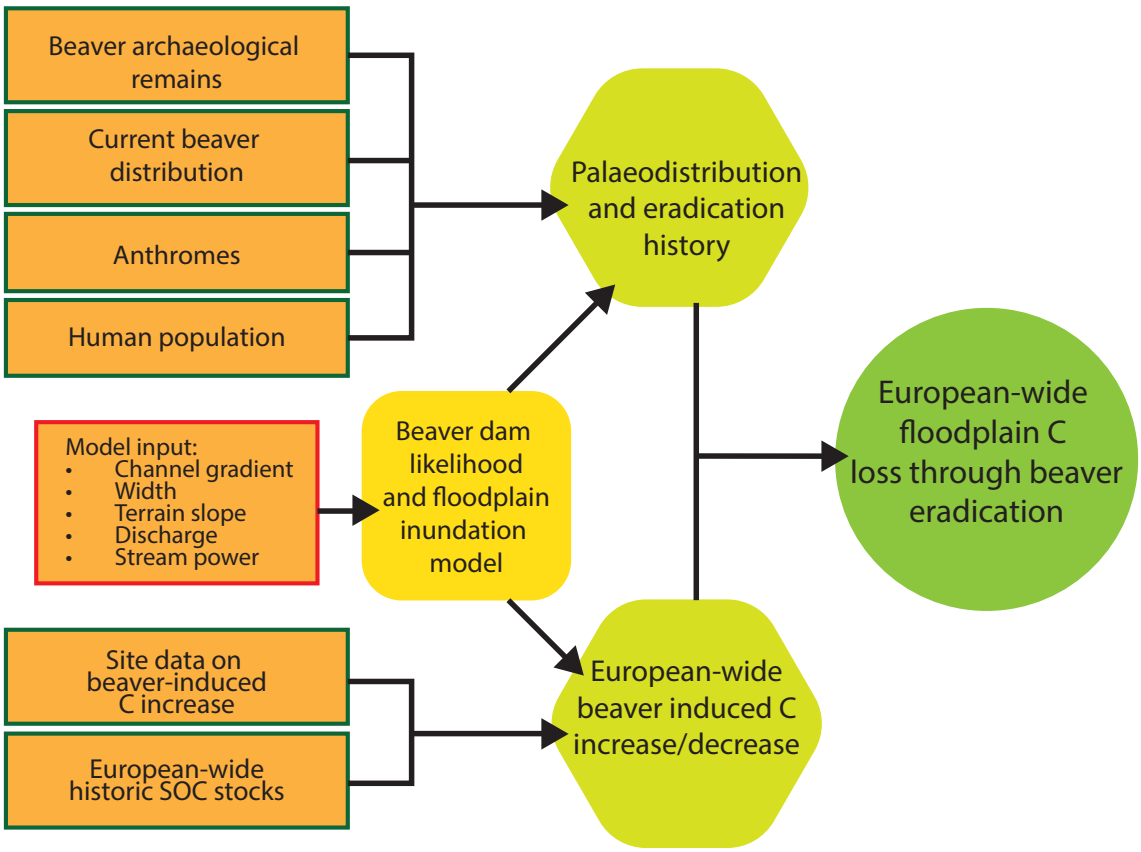


**Figure 6:** Overview of proposed Chapter 2 workflow inputs and outputs. Data (left) outlined in green is available and/or open access; data outlined in red needs to be acquired/generated.

number and density of findings through time. The latter will be done by extracting anthrome type and population density at the location and time period of the point findings, and testing for correlation / applying binomial regression and to assess whether findings can be explained by anthrome or population density through time that suggest (increasing) conflict with humans and intentional eradication. This data will be analysed at the temporal scale of the HYDE 3.3 dataset- in increments of 1000 years between the years 10,000 BCE and 1 CE, and subsequently in increments of 100 years from 100 CE to 2000 CE. This analysis will be conducted using R and QGIS software.

2. European-wide beaver dam likelihood, and subsequent floodplain inundation extent at likely dam locations, will be modelled following the approach of Dennis et al. [18] using channel gradient, channel width, terrain slope, discharge and stream power as predictors of dam likelihood. For sites where dam likelihood exceeds a given threshold for presence or absence of beavers, inundation extent will be calculated based on floodplain elevation relative to the water surface and an assumed dam height. These outputs will be used to constrain the potential extent of beaver influence to suitable habitat and floodplain environments within the European-wide datasets.

3. The past extents of beaver influence for the above-mentioned time slices will be overlain onto maps estimating soil organic carbon (SOC) stocks with the same temporal resolution by Sanderman et al. [74]. Site-specific data on beaver-related carbon increase from several sites in Germany and Switzerland [31], and from a meta-analysis of available literature, will be used to determine the relationship between C content for reference and beaver-influenced floodplain sediments. These relationships will then be applied to estimates of palaeo-SOC stocks [74] for the extent calculated to by floodplain inundation model (see previous paragraph) to calculate European-wide C-sequestration potential. Use of reference site C stocks will enable the conditions preceding (hypothetical) beaver occupation and their relative impact on C stock changes [51] to be accounted for. By incorporating data from several sites, and from literature, we intend to represent different environmental variables as much as possible when scaling

up from site-specific floodplain C contents in our modelling approach. Environmental conditions that differ from our site-specific data in the model will be accounted for in the uncertainty of C stock estimates. Combining this C-sequestration potential with beaver palaeodistribution will allow quantification of European-wide (human-induced) floodplain C loss throughout the Holocene, and thus inform potential carbon sequestration measures.

### Chapter 3: Upper reach controls on C: Landslide-induced sediment inputs

Carbon storage and preservation within river systems is closely linked to residence times of sediments and disturbance processes [92]. As such, river systems in active mountain belts with high sediment turnover rates are typically viewed as having lower storage potential than, for example, passive margin systems [53, 81, 34]. However, such high sediment yields [59] contribute disproportionate amounts of C into their rivers [17, 52] of both biospheric and petrogenic (fossil) origin [53, 34, 35, 36]. Landslides play a dual role in both supplying and retaining sediment to and in rivers [45], and the formation of landslide-driven terraces and alluvial flats disrupting the longitudinal profile of a river can form a series of 'storages' in mountain rivers [43, 45]. A build up of landslide debris within the channel can result in conditions enabling the retention of carbon in sediments [43] or of biogenic origin [79] (2), for example through channel and floodplain aggradation, overbank flows, palaeochannel back-filling, burial of floodplain soils and forests, or in the case of full blockage, a landslide-dammed lake [47]. While several studies investigate the C contents of mountain river sediment fluxes to the ocean (e.g. [35, 34, 25], an understanding of the role of landslide sediments in potentially facilitating C storage en route to the ocean is lacking.

Rivers in the western Southern Alps of New Zealand have disproportionately high sediment fluxes on the global scale [59] dominated by landslide derived material [38, 47]. These characteristics, and extensive landslide inventories for this area [43, 42, 44], are well-suited for investigating landslide influence on river morphology and floodplain carbon storage. Alpine rivers in Westland experience frequent disturbance by landsliding (with landslides affecting 11% of channels in the South-Westland region), while the preservation of pre-historic landslide deposits indicate the potential for medium - long term storage within or upstream of channel-blocking landslides [43, 46]. Discrete stages of geomorphic response to landslide blockage of the channel can create storage zones through various processes (e.g. aggradation, overbank deposition, back-filling of cut-off channels or backwater pools, or burial of floodplain forests and soils [47]). This chapter aims **to investigate the effects of (former) landslide dams on channel morphology and the creation of C storage zones**. Although landslide dams completely blocking channels in this region tend to be ephemeral [43], subsequent occlusion or diversion of the channel following dam breach continue to have long-term effects on sedimentation regimes [45, 46], and warrant further research. Landslide-river interactions investigated by Korup [43] and classified into geomorphic impact classes shown in Fig. 7, with 43.5% of these landslides involving occlusion, blockage, or obliteration of the river channel.

| Geomorphic impact class | | Diagnostic landform assemblage |
|---|---|---|
| I | Buffered | No discernible physical contact between landslide toe and river channel. |
| II | Riparian | Landslide slightly protruding or impinging on river channel; concave landslide toe; clear dominance of fluvial erosion. |
| III | Occlusion | Diversion of river channel around convex landslide toe (or breached landslide dam), compared to channel course up- and downstream of the contact zone. |
| IV | Blockage | Occurrence of a landslide-dammed lake. |
| V | Obliteration | Complete burial of valley floor; landslide ponds or dammed lakes, stream piracy, reversal, or chaotic drainage. |

**Figure 7:** Geomorphic impact classes for landslides in south-west New Zealand. From Korup [43].

Study site selection will be informed by a combination of (a) the New Zealand landslide dam database [61]; (b) Google Earth satellite imagery; (c) previous research investigating landslide-impacted reaches

and (d) site accessibility. An inventory of landslide-driven alluvial flats (i.e. potential storage zones) will be made prior to preliminary fieldwork/scoping in January 2026, during which one/multiple site(s) will be selected for data collection. The field campaign for data collection at the selected site(s) will take place in early 2027. Remote sensing data will be sourced from existing available databases; digital elevation models (DEM's) and geological maps of the selected study area will be obtained from the Land Information New Zealand (LINZ) Data Service; aerial imagery from the Crown Aerial Film Library, and satellite imagery through Google Earth Engine.

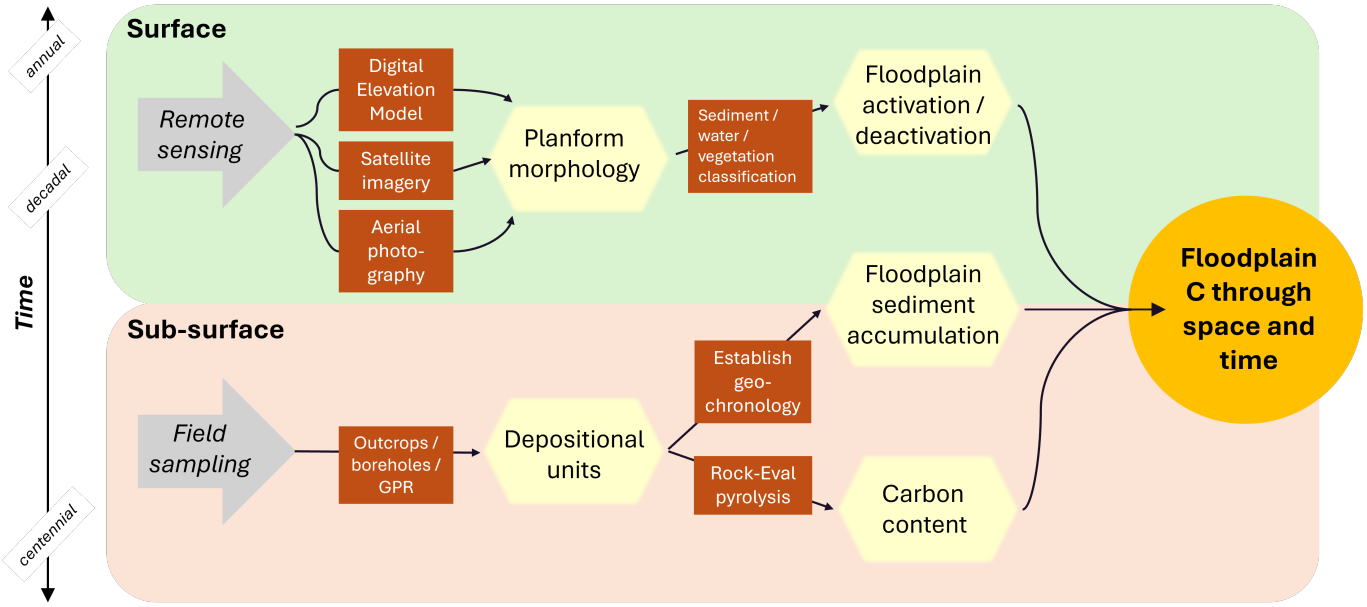


**Figure 8:** Overview of proposed Chapter 3 workflow.

The selected site(s) will be investigated through a combination of methods to link river morphology to floodplain storage through space and time (Fig. 8). Korup [43] found that average channel length directly affected by landslides in south-Westland was 1000 m; thus the spatial scale of the studied reach will be on the order of $10^2 - 10^3$ m. Contemporary processes (annual to decadal time-scale) will be investigated using satellite imagery and/or aerial photography to identify water, sediment, and potential vegetation on the floodplain through time and thus characterise changes in channel morphology and floodplain activity. Floodplain sediment lithologies at the sampled sites will be identified through exposed sections of the floodplain / borehole descriptions (where possible) / ground penetrating radar (**exact methodology to be determined upon field site orientation campaign**), and characterised by the depositional processes responsible for their formation since the time of landslide impact (decadal - centennial time-scale). C contents will be quantified through Rock-Eval pyrolysis, while age control will be provided through optically stimulated luminescence (OSL) and/or radiocarbon dating and/or cosmogenic nuclide dating. A space-for time approach will be used to quantify depositional processes by investigating sites that vary in their time since landslide impact to understand the formation and eventual erosion (i.e. longevity) of landslide-formed alluvial flats. Using a combination of field data and remotely-sensed data for several sites, and linking these to aggradation rates and C content in preserved sediments will enable characterisation of the magnitude and longevity of potential floodplain storage in these landslide affected reaches.

### Chapter 4: Stability of C through time

Studies are increasingly recognising the relevance of assessing SOC stability as well as quantity in the context of carbon budgeting (e.g. [40, 73, 24, 72]). However, these studies are often limited to terrestrial environments. Schmidt et al. [76] describes the persistence of SOM as an ecosystem property, and thus regulated by the physical, chemical, and biological influences of the surrounding

environment. It is therefore relevant to investigate the stability of sequestered C in the context of the environment in which it is found. Although numerous studies quantify floodplain C in absolute terms (e.g. [80, 71, 22, 13]), characterisation of the chemical stability of floodplain carbon related to hydrological, geomorphological or biotic processes remains limited ([27, 97]). This chapter aims **to assess variation in the spatial and temporal stability of floodplain carbon** in the depositional environments investigated in Chapters 1-3. Results of Rock-Eval pyrolysis (see panel) from samples/data obtained in Chapters 1-3 will be analysed to understand how the stability of floodplain C varies between these systems.

**Rock-Eval Pyrolysis**

Rock-Eval pyrolysis, originating in the petroleum exploration industry, is a method used to provide quantitative and qualitative information on the carbon content of rocks and sediments [77, 19, 48]. It is increasingly being applied to soil science due to its simplicity (requiring only 100 mg of sample) and the ability to provide information on organic matter (OM) composition often difficult to obtain using other methods that require isolation of OM prior to measurement [19]. Measured carbon contents using Rock-Eval are highly reproducible and have been found to have excellent correlation with carbon contents produced through elemental analyses [6, 19]. Carbon contents are obtained through pyrolysis of samples in an inert atmosphere ($N_2$) up to $650°C$ and subsequent combustion (oxidation) of residual carbon up to $850°C$. The amount of hydrocarbons, CO and $CO_2$ released between defined temperature limits during pyrolysis and oxidation are plotted as curves (Fig. 9) that enable assessment of the quantity of organic and inorganic carbon within the sample, and of the thermal stability of organic carbon. Parameters calculated through curve integration used to characterise carbon content are shown in Table 1.

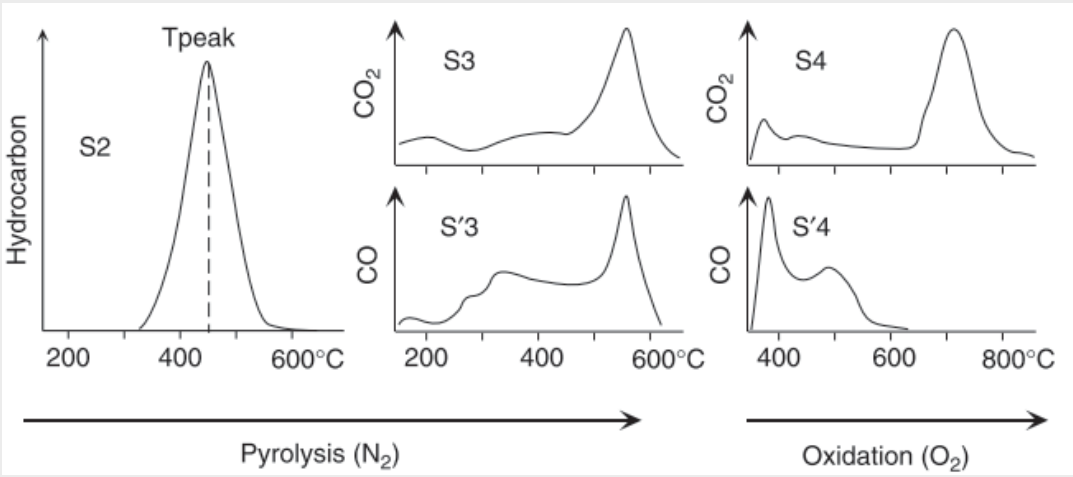


**Figure 9:** Signals measured during Rock-Eval pyrolysis (from Sebag et al. [77]. S1 curve (measuring free hydrocarbons released at $< 200°C$ not shown).

Stability of SOC in sediments depends on four mechanisms: (1) selective preservation of recalcitrant compounds; (2) physical stabilisation by interaction with mineral surfaces; (3) aggregation (spatial inaccessibility to decomposition) and (4) climatic stabilisation e.g. freezing, anaerobic conditions, waterlogging [56, 84, 73]. Rock-Eval pyrolysis has successfully been applied to characterise soil carbon stability (e.g. [73, 4, 20, 75, 78]) and to provide insight into the mechanisms of C preservation. Persistent OC is more thermally stable than labile OC i.e. requires higher temperatures for pyrolysis [4]; thus, the S2 curve (Fig. 9) and a series of parameters calculated based on this curve (Table 1) can be used to describe the thermal characteristics of SOM to indicate its relative stability. In a typical soil profile, degradation/oxidation of labile carbon fractions is reflected in decreasing HI and increasing OI indices with depth [19], i.e. deeper horizons contain higher proportions of stable C. The parameters obtained can also shed light on the sources of OM. For example, bimodal distributions

**Table 1:** Indices calculated based on RockEval pyrolysis and oxidation curves (after Sebag et al. [78], Lafargue et al. [48], Disnar et al. [20]; see also Fig. 9 for explanation of measured signals; HC= hydrocarbon; SOM= soil organic matter).

| Index | Description | Units |
|---|---|---|
| TOC | Total organic carbon (sum HC compounds, CO, $CO_2$) | wt.% (dry soil/sediment) |
| MinC | Mineral carbon content | wt.% (dry soil/sediment) |
| Tmax | Temperature at peak of S2 curve (where max HC's are released) | $°C$ |
| HI | *Hydrogen index*<br>(HC's relative to TOC) | $mg\,HC\,g^{-1}\,TOC$ |
| OI | *Oxygen index*<br>(oxygen released as CO and $CO_2$ relative to TOC) | $mg\,O_2\,g^{-1}\,TOC$ |
| I-index | Indicator of thermally labile SOM ($I$ for *immature*)<br>Calculated as $log_{10}(A_1 + A_2/A_3)$<br>where A is area under the S2 curve between temperatures:<br>$A_1 = 200°C$ - $340°C$ - labile bio-polymers<br>$A_2 = 340°C$ - $400°C$ - resistant bio-polymers<br>$A_3 = 400°C$ - $460°C$ - immature geo-polymers<br>$A_4 = > 400°C$ - refractory geo-polymers | NA |
| R-index | Indicator of thermally stable carbon ($R$ for *refractory*)<br>Calculated as the proportion of area under S2 curve at $> 400°C$ | 0 - 1 |
| R400 | Indicator of OM quality<br>Calculated as the proportion of area under S2 curve at $< 400°C$ | 0 - 1 |

within the S2 pyrolysis curve can reflect OM inputs of different thermal stability [65]; high variability in HI can reflect the contributions of different biogenic material (i.e. litter; [78]); comparison of HI and OI indices can enable separation of aquatic vs. terrestrial OM inputs [11].

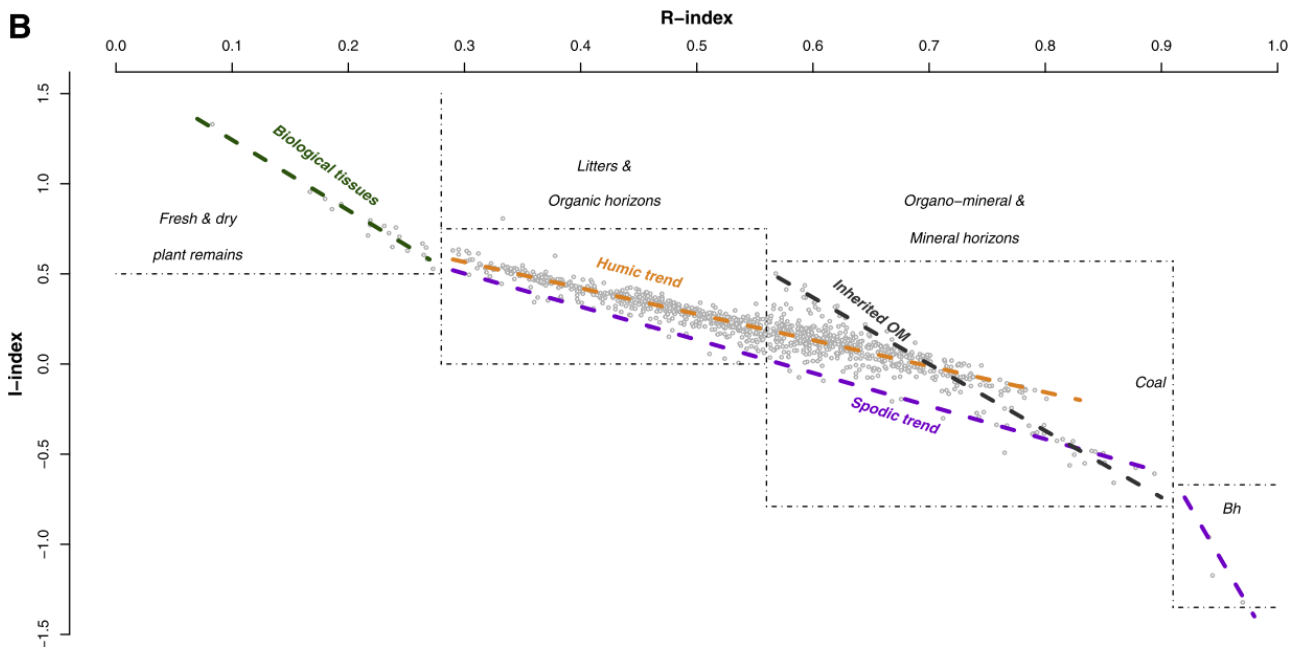


**Figure 10:** Rock-Eval I-R plot from Sebag et al. [78]. Plot delineates boundaries corresponding to different horizons/sample type, and defines linear trends in I-R values for different signatures of OM stabilisation processes. Samples used in classification (grey dots) cover a range of bio-climatic settings and parent materials.

Following the approach of Saenger et al. [73] and Sebag et al. [78], C stocks for samples from each site will be classified in to labile, resistant and refractory C pools, corresponding to areas A1, A2 and A3

integrated under the S2 pyrolysis curve (Fig. 9; Table 1). Sebag et al. [78] defined linear relationships for several SOM stabilisation processes based on comparison of the I and R indices (Table 1; Fig. 10), which can be used to identify mechanisms for C stabilisation; for example in organic horizons stabilisation of C is attributed to decomposition of labile compounds and thus a relative increase in recalcitrant compounds; in mineral horizons higher contributions of stable C to the S2 signal are linked to OM adsorption onto mineral surfaces. Analysis of the obtained parameters (Table 1) will be used to characterise C stability in floodplain sediments and to determine any trends in preservation linked to the respective depositional environments (Chapters 1-3) and processes. The relative contributions of the C pools of varying stability to overall C contents will be calculated (see Saenger et al. [73]), and the I-R indices for each sample will be plotted on the I-R diagram (Fig. 10) to determine whether the identified trends in OM stabilisation hold true for floodplain sediments.

**Summary**

This research will contribute towards a better understanding of floodplain C accumulation and storage by investigating several controls on fluvial depositional within the different process domains [60] making up the longitudinal profile (upper, middle and lower reaches) of river networks. By investigating C storage in floodplain sediments both spatially (through C stock quantification) and temporally (through calculating accumulation rates and stability), we will provide a more comprehensive understanding not only of the (poorly constrained) storage potential in floodplains [5], but also of the stability of floodplain C through time, which is essential to understand from carbon budgeting (and thus climate change mitigation) and nature restoration perspectives.

**Feasibility**

Fig. 11 provides an overview of the project budget and estimated costs. Since the study site and exact methodology for Chapter 3 remain to be determined (see below), costs associated with laboratory analyses, field installation and sampling equipment are not yet constrained (highlighted in blue).

Chapters 1 and 2 of the research proposal are considered low-risk. Chapter 1 makes use of a study site that has been previously investigated; insights from these studies will inform sampling locations and indicate that our proposed methodology using corings and radiocarbon dating will be successful. Logistically, the associated field campaigns are considered feasible since the site is located in the Netherlands, which enables flexibility in planning field visits/extra sampling campaigns (see Timeline in Fig. 12). Chapter 2, being based on analysis of already existing and / or open-access data, is also low-risk regarding methodology and logistics. In terms of time constraint there may be more uncertainty due to collaboration with external parties. However the nature of the research enables flexibility since we are not dependent on sampling campaigns or field seasons, and expected deliverables and timing thereof will be clearly communicated to mitigate this uncertainty as best as possible.

| **Incoming** | **€ / unit** | **Qty.** | **Total** |
|---|---|---|---|
| Laboratory analysis | € 50,620 | 1 | € 50,620 |
| Field Installations | € 4,605 | 1 | € 4,605 |
| Sampling materials | € 800 | 1 | € 800 |
| Travel | € 14,215 | 1 | € 14,215 |
| TSP | € 4,000 | 1 | € 4,000 |
| | | | **€ 74,240** |
| **Costs** | **€ / unit** | **Qty.** | **Subtotal** |
| Radiocarbon dating | € 385 | 30 | € 11,550 |
| OSL dating | € 700 | 28 | € 19,600 |
| | | | |
| Rock-Eval analysis | € 1,500 | 3 | € 4,500 |
| Other analyses | *tbd* | *tbd* | *€ 14,970* |
| | | | **€ 50,620** |
| Field installations | *tbd* | *tbd* | *€ 4,605* |
| | | | **€ 4,605** |
| Sampling materials | *tbd* | *tbd* | *€ 800* |
| | | | **€ 800** |
| Fieldwork NZ 1 | € 2,608 | 1 | € 2,608 |
| Fieldwork NZ 2 | € 9,955 | 1 | € 9,955 |
| Fieldwork NL | € 1,653 | 1 | € 1,653 |
| | | | **€ 14,215** |
| Courses | € 1,415 | 1 | € 1,415 |
| Conferences | € 1,560 | 1 | € 1,560 |
| TSP other | | | € 1,025 |
| | | | **€ 4,000** |
| | | | **€ 74,240** |

**Figure 11:** Incoming budget and estimated costs for the project (tbd= to be determined)

Chapter 3 contains the most uncertainty given that the study site, and thus exact methodology, is yet to be determined, as well as through logistical and financial constraints associated with planning field

campaigns in New Zealand. An orientation field campaign is planned in January 2026 (coinciding with a visit for an international conference) to identify a suitable study site and engage with local scientists to utilise their expertise/recommendations, which is intended to mitigate methodological uncertainty in the actual sampling campaign taking place the following year (Fig. 12). Logistically, much of the uncertainty associated with doing fieldwork in a foreign country is reduced because of my connections in and familiarity with NZ, it being my country of origin. Financially, these campaigns are estimated to remain within the project budget.

Chapter 4, making use of the samples collected in Chapters 1-3, does not involve any risks not mentioned above. Laboratory equipment is available at WUR for analysis of samples, and agreements for use of equipment and associated costs (presently underway) do not present any foreseen risks.

**Timeline**

Fig. 12 shows the proposed timeline for the PhD research.

## 2.6 Data Management

Data generated during this research will be stored on OneDrive and will be back up on the university (W:) drive. Upon publication of any manuscripts, data will be made openly available via the 4TU.ResearchData repository. The data is owned by Wageningen University Research. Full details on data management are available in the data management plan submitted to the Graduate school for Production Ecology & Resource Conservation in September 2025.

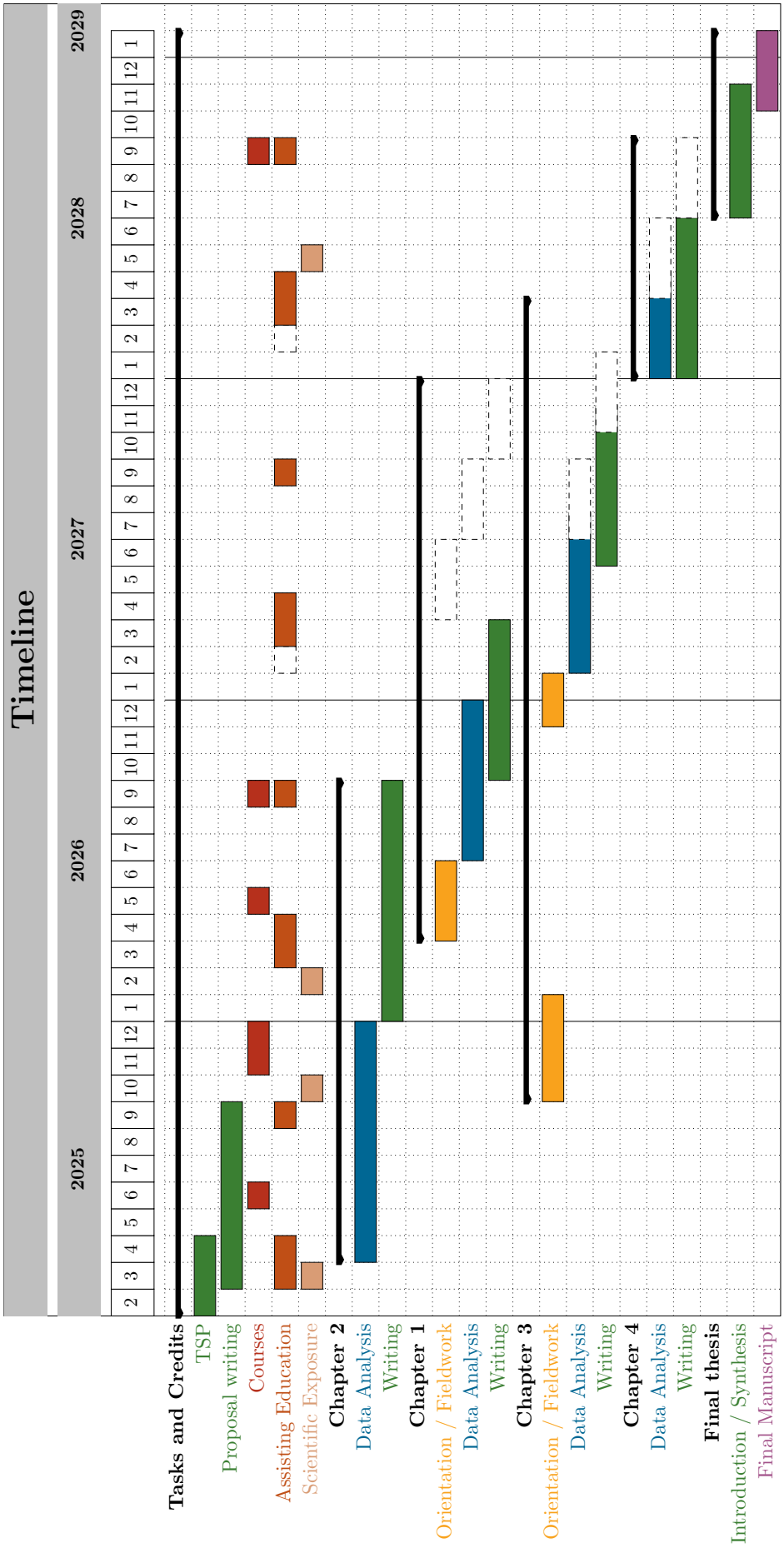


**Figure 12:** Proposed timeline for PhD research

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